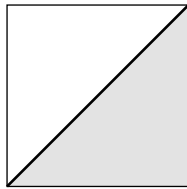


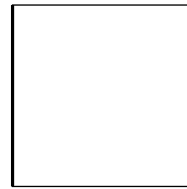
Use whole squares inside the gray shape. Cover all squares with some gray area. A touch at an edge or corner adds no square.



whole

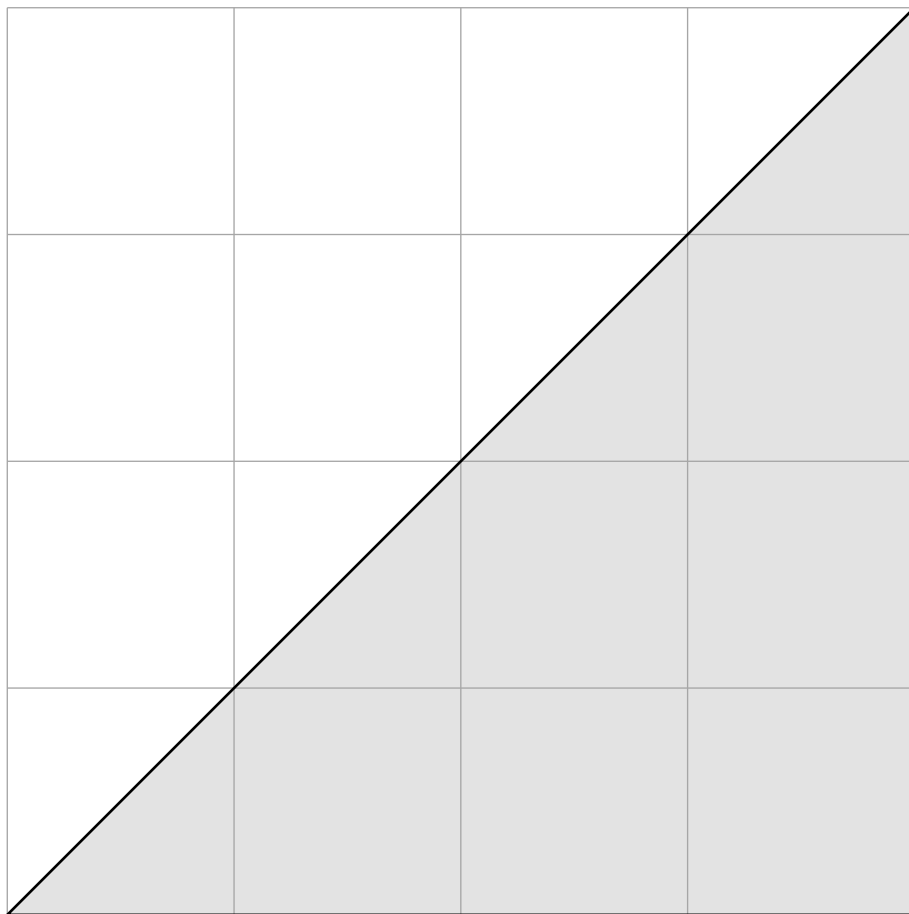


partial

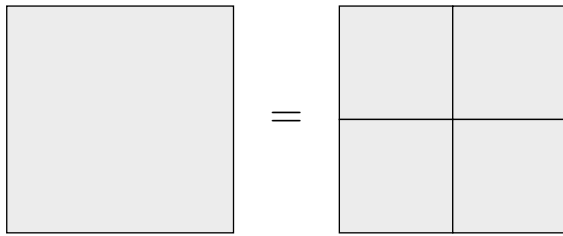


outside

Problem 1: Mark all the whole squares inside the triangle. Cover the triangle with as few grid-square tiles as you can.

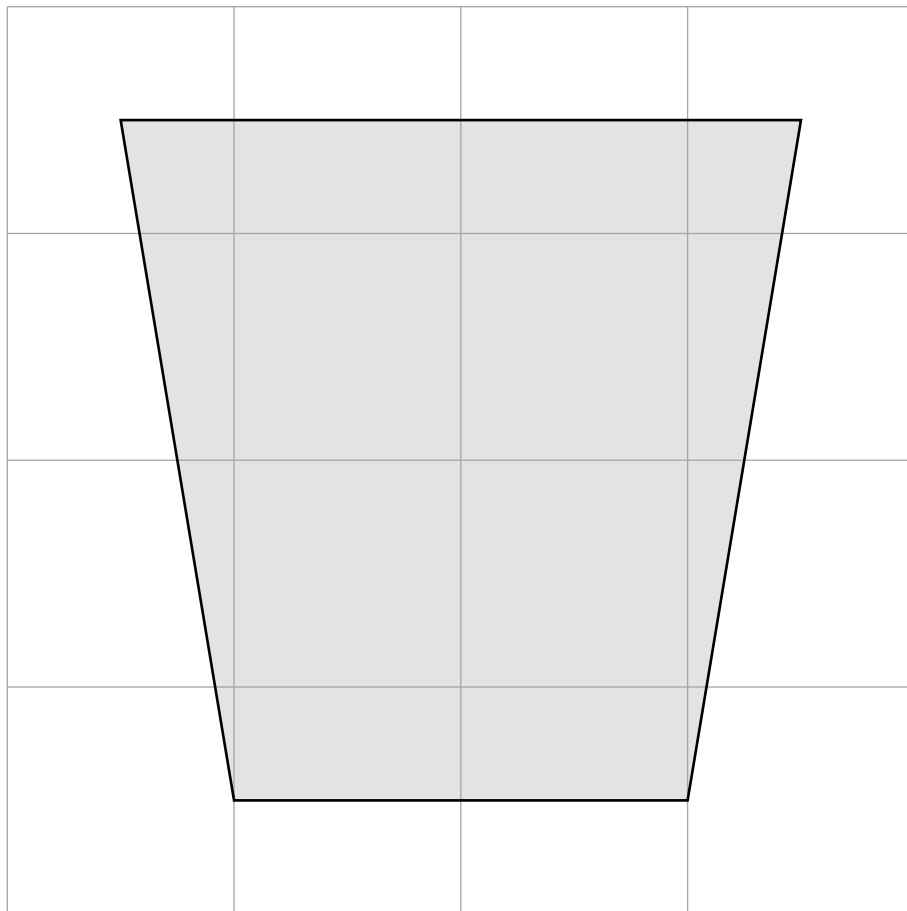


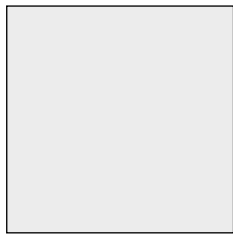
Inside: _____ big squares Cover: _____ big squares



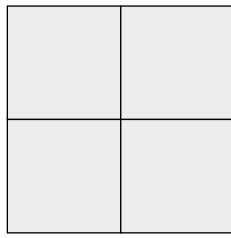
1 big square
= 4 small squares

Problem 2: Cover this shape, then split two big tiles into four small tiles each. Choose the two big tiles that let you remove the most small tiles while keeping the gray shape covered.



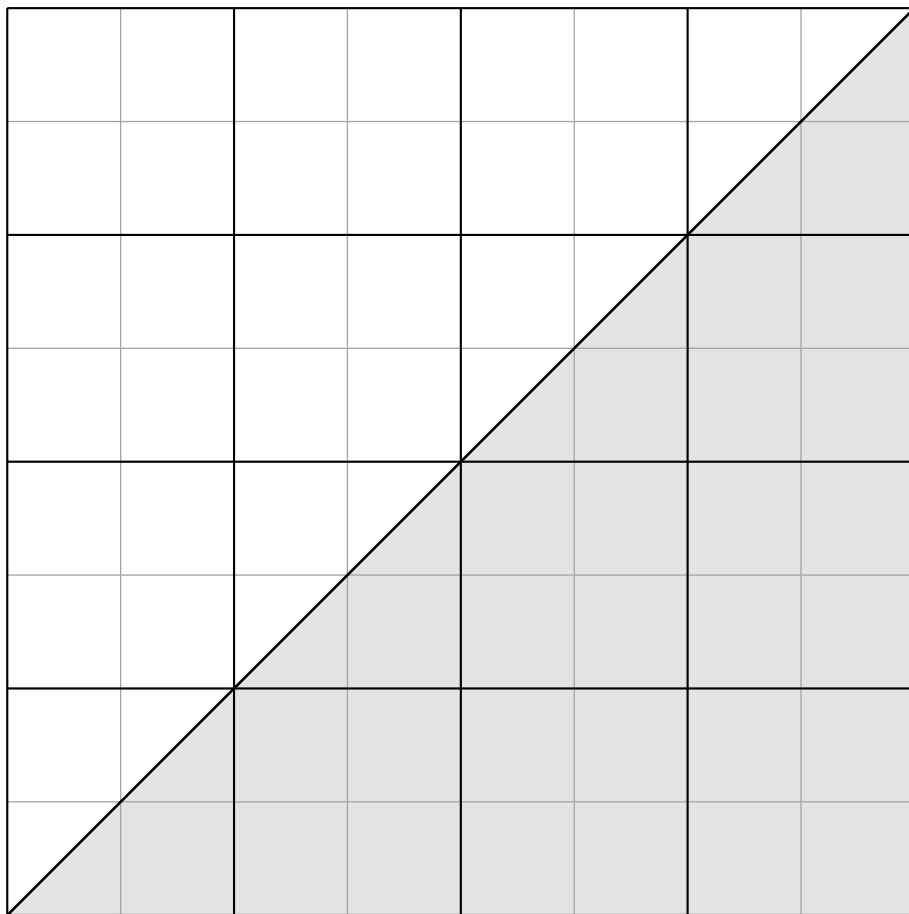


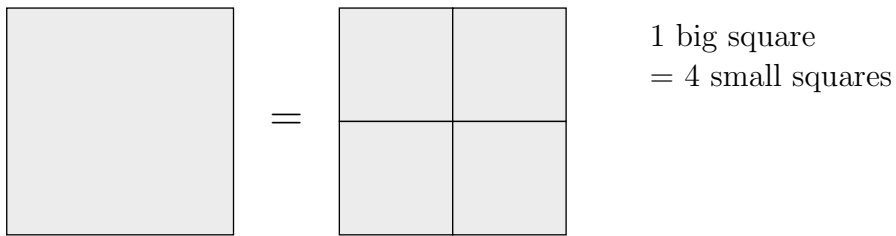
=



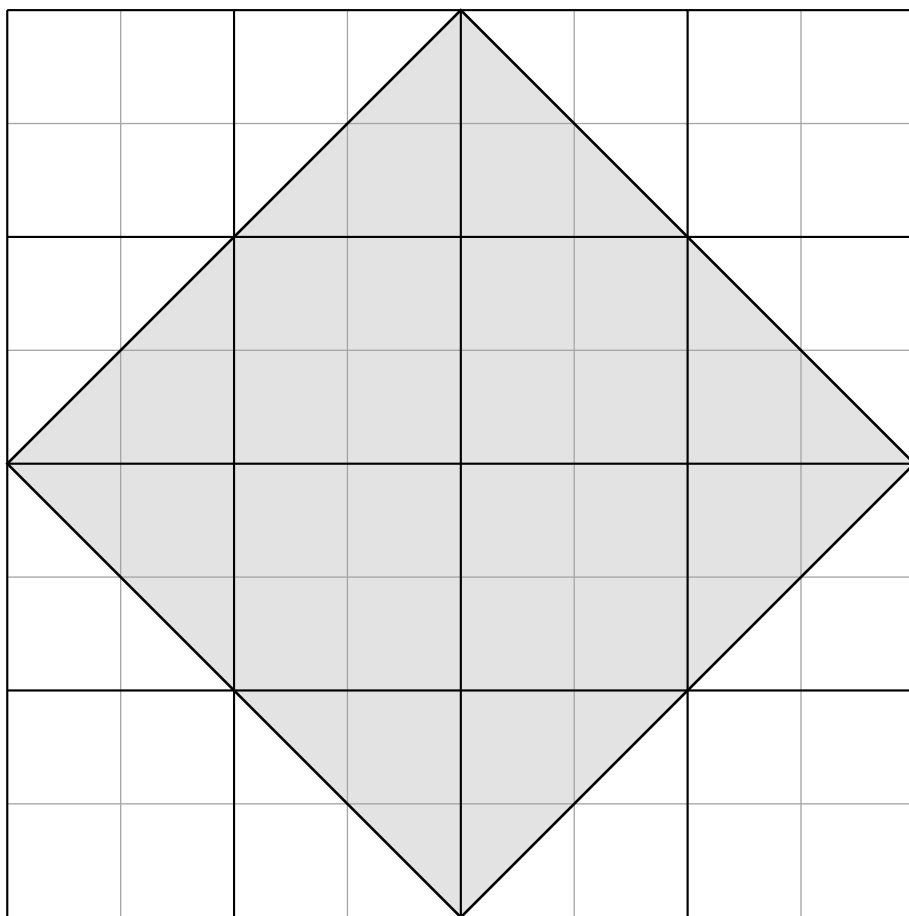
1 big square
= 4 small squares

Problem 3: Use small squares to get more area inside the triangle and less area outside it in your cover. Show which pieces change from the big-square covers.

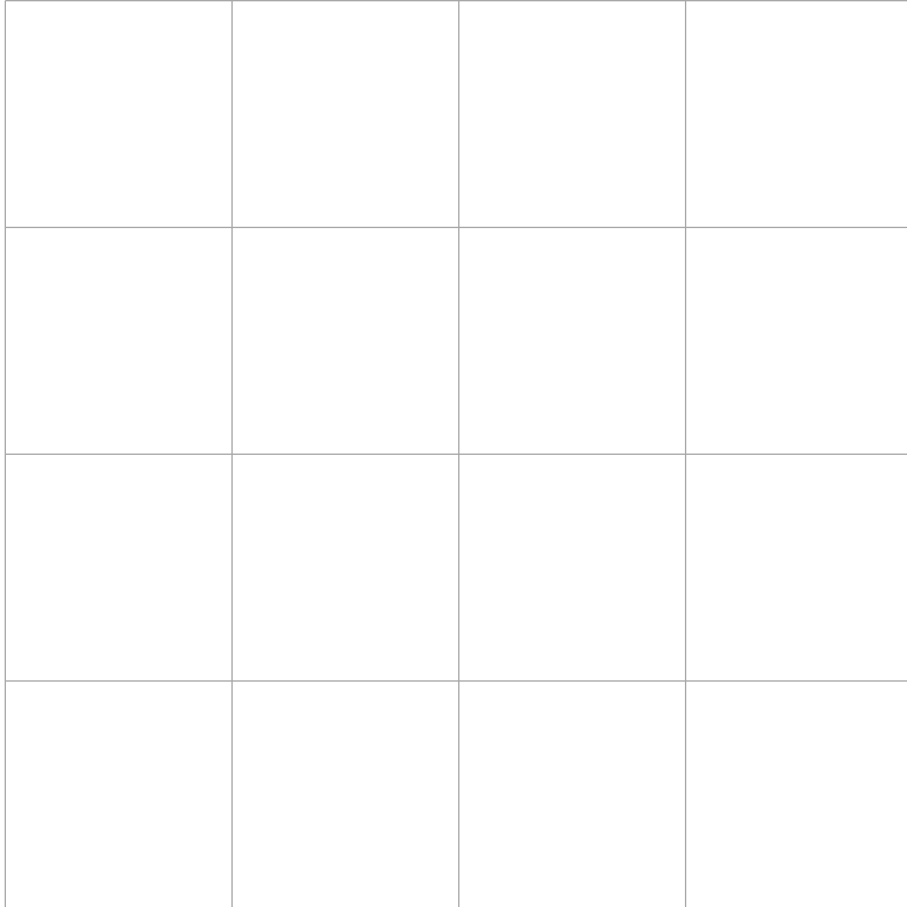




Problem 4: Change the diamond so four outside small squares become partly gray. Keep every big square whole, partial, or outside as it was.



Problem 5: Draw two shapes, one at a time, each with 3 whole and 3 partly gray squares. Make one with less than 4 big squares of gray and one with more than 5.



Problem 6: Keep the same whole and partial squares. Can you make the larger shape bigger again without filling another whole square?

One grid square is 1 big-square unit.